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MoEActok: A MoE-based Action Tokenizer for Vision-Language-Action Models

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Abstract

*Recent works on vision-language-action (VLA) models have made great progress in exploring action tokenizers that convert continuous control signals into discrete tokens to align with LLM/VLM training paradigms. These approaches typically train a single tokenizer over entire manipulation trajectories, which often comprise multiple distinct skills and thus pose a challenging optimization trade-off. To address this issue, we introduce **MoEActok**, a novel **act**ion **tok**enizer that employs a mixture-of-experts (**MoE**) quantizer to produce skill-aware discrete representations for VLA models. MoEActok utilizes a clustering-driven MoE VQ-VAE mechanism in which each expert specializes in a particular skill. The key components are: (a) an action-skill decoupling strategy that uses k-means clustering to group action chunks, aligning clusters having similar skills; (b) a skill-aware training paradigm that augments VLA models with skill-conditioned context, improving skill grounding; and (c) an adapter that projects shared encoder representations into skill-specific latent spaces for specialized quantization, and subsequently harmonize the heterogeneous quantized representations back into a unified space for coherent reconstruction by the shared decoder. We evaluate MoEActok-based VLA models against multiple prior action tokenizer baselines in the RoboTwin and Simpler-Env simulators, and further assess zero-shot transfer on three real-world tasks. Across both simulated and real-world settings, MoEActok-based VLA substantially outperforms existing discrete tokenization methods. Code is available at https://github.com/cpaaax/MoEActok.*

1. Introduction

Vision-language-action (VLA) models [2, 3, 5, 11, 12, 17, 23, 26, 30, 32, 35] have revolutionized robotic manipulation by integrating multimodal inputs to generate executable action sequences, drawing on the success of large-scale pre-training in language and vision domains [1, 9]. These models typically leverage autoregressive architectures [28] to predict latent discrete tokens, with or without a separate action decoder, that represent continuous robot actions, enabling efficient training and inference similar to those in large-language-models (LLMs) and large vision-language-models (VLMs). However, the effectiveness of such models relies on quality of action tokenization process, which distills high-dimensional, temporally coherent control signals into compact, semantically rich discrete representations.

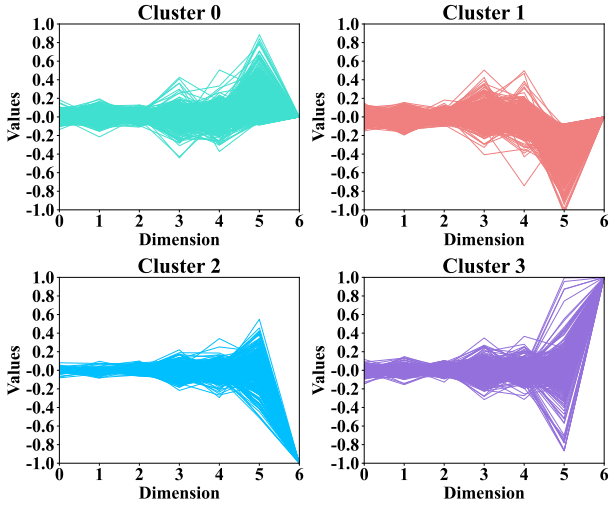


Figure 1. Clustering results of actions on BridgeData V2. The seven dimensions (0-6) represent the robot end-effector actions: x, y, z coordinates for position, roll, pitch, yaw for orientation, and gripper action.

Early VLA approaches [3, 11, 35] rely on straightforward discretization techniques, such as uniform binning of individual action dimensions, which often struggle with the nuances of dexterous tasks due to their inability to capture inter-step dependencies. Recently, more advanced methods have introduced structured tokenization paradigms to mitigate the issue. For instance, FAST tokenizer [23] proposes frequency-domain transformations followed by byte-pair encoding to facilitate autoregressive prediction in challenging manipulation scenarios. Complementing this, residual

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VQ-VAE frameworks (*i.e.*, VQ-BET [12], VQ-VLA [31]) employ multiple residual vector quantization layers to hierarchically transform action sequences into discrete tokens.

Despite these notable advances in tokenization design for VLAs, a fundamental challenge persists that these tokenizers are trained holistically across full trajectories that integrate diverse and heterogeneous skills, such as coordinated movement and precise grasping. These approaches, while intuitively appealing for capturing end-to-end behaviors, inadvertently introduce representation conflicts as they require the model to learn from conflated signals. For example, in robotic manipulation tasks, a single trajectory might seamlessly blend translational locomotion with fine-grained end-effector adjustments, imposing competing demands on the tokenizer's capacity to process actions of different skills. As reflected by the clustering phenomenon in Figure 1 for action chunks from BridgeData V2 [29], this heterogeneity shows in distinct yet intermixed specialization patterns: clusters 0 and 1 predominantly capture movement primitives, with cluster 0 and cluster 1 corresponding to opposite rotations around the z-axis; in contrast, clusters 2 and 3 focus on grasping actions, where cluster 2 aligns with gripper-closing motions and cluster 3 with gripper-opening motions. This mixed-signal optimization necessitates a compromise in representation fidelity, as a unified quantizer struggles to specialize in the distinct kinematic patterns and temporal scales of individual skills, resulting in suboptimal generalization. Moreover, without explicit mechanisms to disentangle skill-specific structures, downstream VLAs may underperform in grounding observations to precise action primitives.

To address these challenges, we propose **MoEActok**, a mixture-of-experts (MoE) [7] action tokenizer that generates skill-specialized discrete tokens tailored for VLAs. Individual experts are specialized via k-means partitioning of action chunks into semantically coherent clusters. This ensures high intra-cluster similarity aligned with distinct skills, such as rotational movements or grasping primitives, while enabling dynamic routing to maintain representational compactness and fidelity. Complementing the model design, we propose an adapter that bridges shared and skill-specific latent spaces before and after quantization, enabling effective cross-skill representation learning. To further enhance skill grounding, we incorporate a skill-aware training paradigm that conditions VLA models on skill clusters, enabling hierarchical prediction from coarse-grained skill categories to fine-grained action tokens. This strategy mitigates optimization conflicts, promotes modular generalization, and improves the alignment of observations with precise action primitives.

To validate the effectiveness of the proposed method, we conduct experiments on two widely used simulators, including RoboTwin [6] and Simpler-Env [14]. Moreover, we perform zero-shot experiments with real robots to evaluate the effectiveness of using VLAs trained with collected data from RoboTwin, directly on tasks in the real world. In summary, our work makes the following key contributions:

- We introduce MoEActok, a novel mixture-of-experts tokenizer that decomposes action discretization into skill-specific quantization, where each expert specializes in encoding actions from different behavioral clusters identified through unsupervised learning clustering.
- We propose a skill-aware training paradigm that can augment VLAs with coarse-to-fine action prediction.
- We develop an adapter that aligns representations between the unified encoder-decoder architecture and specialized expert quantizers, facilitating coherent action reconstruction.
- We carried out comprehensive experiments in both simulated and zero-shot real-world settings to showcase the effectiveness of the proposed approach.

2. Related Work

2.1. Vision-Language-Action Models

Vision-language-action (VLA) models have rapidly evolved into a cornerstone for endowing robots with versatile manipulation capabilities, bridging the gap between high-level semantic understanding and low-level physical execution [2, 3, 5, 21, 25, 32, 35]. By leveraging the pre-trained representations from large language models (LLMs) and vision-language models (VLMs), these architectures enable robots to interpret natural language instructions alongside visual inputs, generating coherent action sequences for complex tasks. Pioneering efforts like CLIPort [25] and PerAct [21] laid the groundwork by fusing CLIP-encoded vision with modular policy networks to handle language-guided object interactions. Subsequent advancements in the RT family [2, 3, 35] scaled this paradigm through massive internet-scale pre-training, introducing discrete action representations that facilitate efficient fine-tuning on robot-specific data. Building on these foundations, contemporary VLA systems emphasize broad generalization and multitask proficiency. For instance, Octo [26] introduced an expansive dataset spanning diverse robot embodiments, fostering policies that adapt across varied hardware setups. Similarly, OpenVLA [11] demonstrated robust zero-shot transfer to real-world household scenarios by distilling VLMs into compact action predictors.

2.2. Action Tokenization

Central to the success of these autoregressive VLAs is the design of action tokenization schemes, which transform raw, multidimensional trajectories into discrete sequences amenable to next-token prediction [24]. Traditional approaches often rely on uniform binning, discretizing each

action dimension independently at every timestep [3, 11, 35]. While computationally lightweight, this method struggles with temporal correlations in high-dimensional spaces, leading to redundant tokens and suboptimal learning, particularly for smooth, predictive motion patterns. To address these limitations, researchers have turned to more sophisticated compression techniques that exploit the inherent structure of action signals. Vector quantization (VQ) emerges as a promising direction, enabling latent space clustering for compact representations [12, 31]. For example, VQ-VLA [31] advances this by training convolutional residual VQ-VAEs on scaled datasets, blending real-world noisy trajectories with vast synthetic simulations to yield tokenizers that enhance long-horizon planning. Complementing this, FAST [23] draws from signal processing roots, applying discrete cosine transforms (DCT) to shift actions into a frequency domain before employing byte-pair encoding (BPE) for iterative merging of correlated coefficients. However, none of these works take into consideration decoupling heterogeneous action skills, such as precise grasping versus broad movement, into specialized representations, often resulting in entangled latent spaces that hinder fine-grained control. In this work, we address this gap by introducing a novel tokenizer that leverages multiple experts to modularly learn and disentangle diverse skill distributions, enabling more adaptive action generation for diverse robotic tasks. Several recent works bypass action tokenization entirely by generating continuous actions through alternative decoding strategies, such as diffusion heads [4, 8, 10, 13, 15, 16, 18, 19, 22]. While effective, the majority of these approaches require architectural modifications incompatible with standard autoregressive transformers. In contrast, our tokenization-based method seamlessly integrates with existing LLMs, preserving their architectural benefits without structural changes.

3. Method

3.1. Problem Formulation

In this work, we explore a comprehensive robotic manipulation dataset denoted as $D = \{\{(o_1, s_1, a_1), ..., (o_{T_i}, s_{T_i}, a_{T_i}), L_{m,i}\}_{i=1}^{N_m}\}_{m=1}^{M}$, which consists of M distinct manipulation scenarios. In the mth scenario, there are N_m demonstration trajectories. The ith trajectory spans T_i timesteps, where at each timestep t, we record: an RGB frame o_t capturing the environmental view, the robot's proprioceptive state s_t, and the executed control action a_t. To elaborate on the details of a_t, we take the actions in BridgeDataV2 [29] as an example. The robot arm operates with 7 Degrees of Freedom (DoF), and $a_t \in \mathbb{R}^7$ is represented by a 7-dimensional vector. In this vector, the first six dimensions correspond to the positional coordinates (x, y, z) and the rotational angles (roll, pitch, yaw). The final dimension pertains to the gripper's action, where a value of 1 indicates an open action and a value of 0 means a closed action. Accompanying each sequence is a concise natural language sentence $L_{m,i}$ that describes the objective of the scenario. The overarching aim is to derive an end-to-end policy $\pi_\phi(a_{t:t+k-1}|o_t, s_t, l)$ capable of generating k-step action chunks directly from visual observation, proprioceptive states, and instruction text.

3.2. Action-Skill Decoupling

MoEActok training relies on a mixture-of-experts architecture, where each expert focuses on one group of actions. Therefore, partitioning action chunks across trajectories into semantically distinct groups is necessary. As supported by the action clustering results shown in Figure 1, actions exhibit clear clustering patterns based on the seven dimensions of actions, demonstrating that actions naturally form distinct groups. Importantly, such kinematic-based grouping is more conducive to model learning and generalization, as it reflects the underlying structure of manipulation primitives. Therefore, we propose an efficient, unsupervised clustering procedure that automatically categorizes action chunks based on their inherent features.

For an input action chunk $a_{t:t+k-1} \in \mathbb{R}^{k\times 7}$, we derive a compact global representation $\bar{a} \in \mathbb{R}^7$ that encapsulates the chunk's aggregate dynamics. The initial 6 dimensions aggregate the cumulative arm motion via element-wise summation: $\bar{a}_{1:6} = \sum_{j=t}^{t+k-1} a_{j,1:6}$. Meanwhile, the seventh dimension quantifies net gripper displacement as the endpoint difference: $\bar{a}_7 = a_{t+k-1,7} - a_{t,7}$. This design ensures that the first six dimensions of $\bar{a}$ encapsulate the aggregate arm kinematics across the chunk. In contrast, the seventh dimension isolates the gripper action, thereby disentangling translational and end-effector primitives. We then apply the k-means clustering algorithm to these derived global chunk representations, yielding K distinct cluster centroids. Each cluster is interpreted as a specialized skill category.

3.3. MoEActok Tokenizer

Building upon the action-skill decoupling framework established above, we introduce MoEActok, a novel mixture-of-experts tokenizer that leverages the identified skill clusters to achieve more effective action representation learning. Our tokenization architecture is fundamentally grounded in vector quantization principles, extending variational autoencoder methodologies [27] with a multi-expert paradigm tailored for heterogeneous skill distributions. The core architecture consists of three interconnected components: a shared encoding module, a mixture-of-experts quantization layer, and a unified decoding module. The encoder and decoder employ a hybrid architectural design that synergistically combines convolutional neural networks with multi-head transformer layers, enabling both local tempo-

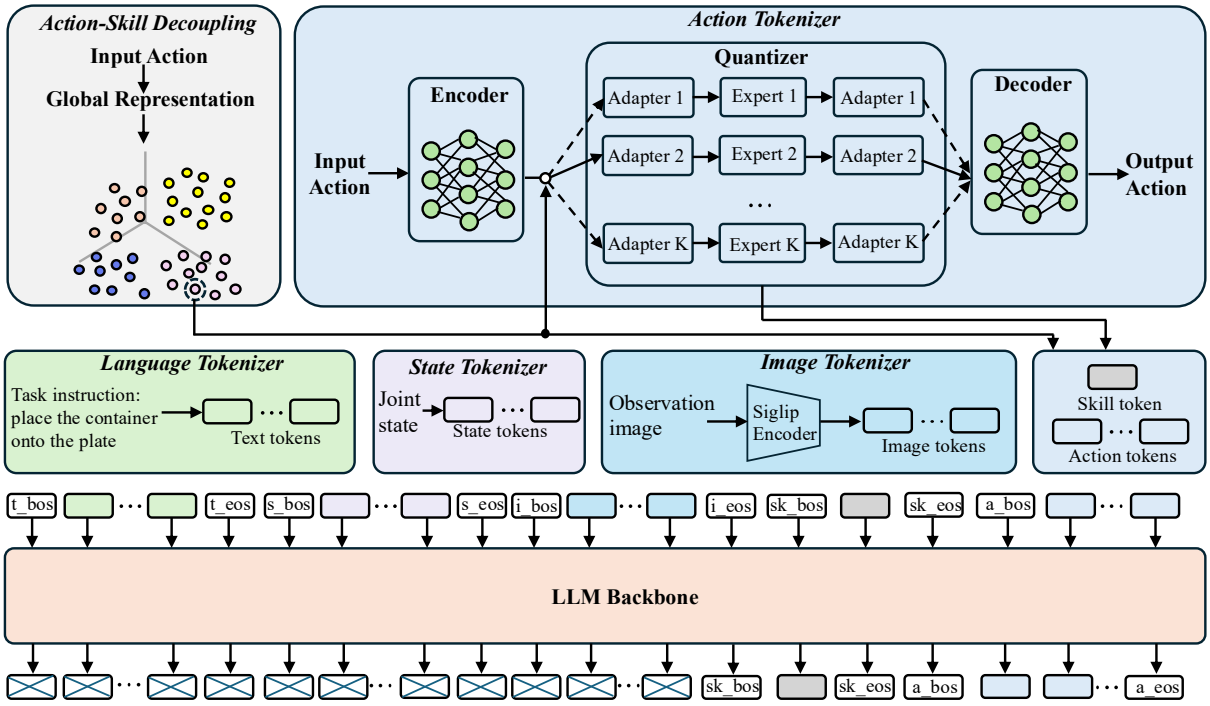


Figure 2. Overview of MoEActok action tokenizer (top) and the MoEActok-based VLA model (bottom).

ral pattern extraction and long-range dependency modeling. Specifically, the encoder transforms input action chunks $a_{t:t+k-1}$ into compressed latent representations z that capture essential behavioral patterns.

The key innovation lies in our mixture-of-experts quantization module, which addresses the fundamental limitation that a single quantizer struggles to optimally represent the diverse distributional characteristics inherent across different skill categories. Motivated by our K-cluster action-skill decomposition, we deploy K specialized vector quantizers $\{VQ_1, VQ_2, ..., VQ_K\}$ where each quantizer VQ_i maintains its own codebooks. During quantization, the skill category h derived from the clustering procedure directs the latent representation z to the related expert quantizer, ensuring that each skill-specific quantizer can specialize in capturing the nuanced patterns within its designated action category. Formally, given a latent representation z and its associated skill category $h \in \{1, 2, ..., K\}$, z is transformed into discrete tokens through the quantization process as follows:

$$z_q, q = \arg\min_{c \in VQ_h} ||z - c||_2, \tag{1}$$

where q is the quantized index, z_q is the corresponding embedding vector from the related codebook of VQ_h.

z_q is subsequently processed by the decoder to reconstruct the original action chunk $a_{t:t+k-1}$. This design ensures that each quantizer can develop specialized representations optimized for its respective skill domain.

Furthermore, given that each quantizer specializes in distinct action skill clusters with potentially disparate latent distributions, we introduce a skill-adaptive mapping mechanism to ensure optimal feature alignment across the heterogeneous expert domains. This mechanism comprises two complementary components: a pre-quantization adapter $\boldsymbol{A}_h^{pre}$ and a post-quantization adapter $\boldsymbol{A}_h^{post}$. The pre-quantization adapter $\boldsymbol{A}_h^{pre}$ transforms the shared encoder representation z into a skill-specific latent space tailored for quantizer VQ_h:

$$z' = \boldsymbol{A}_h^{pre}(z) = W_1(\sigma_1(W_2(z) * W_3(z)) + \sigma_2(W_4(z)) \tag{2}$$

where W_1, W_2, W_3, W_4 denote trainable linear transformation matrices, σ_1 and σ_2 are the ReLU activation function.

This transformation enables each expert quantizer to operate within its specialized representational subspace, where the codebook embeddings are optimally positioned to capture skill-specific behavioral patterns. The skill-specific latent representation z' is then quantized using the corresponding expert quantizer to obtain the discrete index q and the retrieved quantized embedding z_q.

Conversely, the post-quantization adapter $\boldsymbol{A}_h^{post}$, which employs an identical architectural structure as its pre-quantization adapter, maps the z_q back to a unified latent space compatible with the shared decoder:

$$z_q' = \boldsymbol{A}_h^{post}(z_q) \tag{3}$$

where z_q' is sent to the decoder for reconstruction.

To optimize our MoEActok tokenizer, we follow VQ-VAE [27] and adopt a multi-objective training paradigm that combines three complementary loss components. The

primary objective is the reconstruction fidelity loss $\mathcal{L}_{rec}$, which enforces semantic consistency between the original input action chunks $a_{t:t+k-1}$ and their decoded action $\hat{a}_{t:t+k-1}$:

$$\mathcal{L}_{\text{rec}} = \|\hat{a}_{t:t+k-1} - a_{t:t+k-1}\|_2^2 \tag{4}$$

This loss ensures that the entire encode-quantize-decode pipeline preserves the essential kinematic information necessary for faithful action reproduction.

The second component is the codebook optimization loss $\mathcal{L}_{emb}$, which facilitates the learning of meaningful discrete representations within each expert's codebook. This loss minimizes the distance between the skill-adapted encoder output z' and its nearest quantized embedding z_q:

$$\mathcal{L}_{\text{emb}} = \|sg[z'] - z_q\|_2^2 \tag{5}$$

where $sg[\cdot]$ denotes the stop-gradient operator, ensuring that gradients flow only to the codebook parameters and not back through the encoder during this optimization step.

The third component is the encoder commitment loss L_{com}, which stabilizes the training dynamics by encouraging the encoder to produce representations that align closely with the selected codebook vectors, thereby mitigating erratic switching between different discrete codes:

$$\mathcal{L}_{\text{com}} = \|z' - sg[z_q]\|_2^2 \tag{6}$$

The overall training objective combines these three components with the weighting factors:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \alpha\mathcal{L}_{\text{emb}} + \beta\mathcal{L}_{\text{com}} \tag{7}$$

where α and β are hyperparameters that balance the relative importance of codebook learning and encoder stability, respectively.

3.4. Skill-Aware VLA Training

Our objective is to develop a unified vision-language-action policy $\pi_\phi(a_{t:t+k-1}|o_t, s_t, l)$ that can seamlessly integrate multimodal inputs to generate coherent action sequences for different manipulation tasks.

To enable effective cross-modal learning within a single transformer architecture, we establish a comprehensive tokenization pipeline that converts each input modality into discrete token sequences while preserving semantic integrity. For linguistic instruction l, we leverage the LLM's native tokenization scheme to obtain textual tokens. Proprioceptive state s_t is discretized into 256 bins per state dimension following FAST [23]. Visual observation o_t is processed through SigLIP-SO400M [34], yielding 256 continuous visual tokens that encode spatial-semantic information. Action sequences are tokenized to discrete tokens $\{q_r\}_{r=1}^R$ using our proposed MoEActok, transforming continuous action chunks into discrete action tokens. To maintain clear modality boundaries, we introduce special delimiter tokens, where t_bos/t_eos for textual segments, s_bos/s_eos for proprioceptive state information, i_bos/i_eos for image content, and a_bos/a_eos for action sequences.

Given the multimodal tokenized inputs, the standard VLA training follows an autoregressive objective using cross-entropy loss:

$$\mathcal{L}_{\text{VLA}} = -\sum_{r=1}^{R} \log P(q_r|q_{<r}, o_t, s_t, l) \tag{8}$$

where q_r represents the r-th action token, $q_{<r}$ denotes the preceding action token sequence.

However, this conventional training approach presents a fundamental challenge. The action tokens originate from distinct skill clusters identified during our action-skill decoupling procedure, yet the model needs to implicitly infer the appropriate skill category and generate corresponding tokens during inference without explicit skill guidance. This implicit skill reasoning significantly complicates the learning process, as the model must simultaneously solve skill classification and action prediction tasks within a single objective. To address this limitation, we propose a skill-aware VLA training paradigm that explicitly incorporates the skill clusters $h \in \{1, 2, ..., K\}$ obtained from our unsupervised clustering procedure. This approach decomposes the action generation process into two interpretable stages, i.e., skill identification followed by skill-conditioned action prediction.

The enhanced training objective is reformulated as:

$$\begin{aligned}\mathcal{L}_{\text{VLA}} = &- \log P(h \mid o_t, s_t, l) \\ &- \sum_{r=1}^{R} \log P(q_r \mid q_{<r}, o_t, s_t, l, h)\end{aligned} \tag{9}$$

The first term enforces explicit skill classification capability, requiring the model to predict the appropriate skill cluster h based on the current context. The second term conditions action token generation on both the contextual inputs and the predicted skill category, enabling the model to leverage skill-specific behavioral patterns learned by the corresponding expert quantizer.

4. Experiments

We evaluate our method on two simulation environments, RoboTwin [6] and Simpler-Env [14], as well as on a real-world AgileX Cobot Magic robot.

4.1. Simulation Benchmarks

RoboTwin. RoboTwin is a comprehensive bimanual manipulation benchmark designed to facilitate scalable dataset

Table 1. Success rates of different VLAs across RoboTwin simulation 12 tasks.

Tokenizer	Beat Block Hammer	Move Playingcard Away	Pick Diverse Bottles	Move Can Pot	Move Pillbottle Pad	Click Bell	Handover Mic
Binning	0.11	0.01	**0.27**	0.02	0.00	0.67	0.85
FAST	0.00	0.18	0.06	0.08	0.01	0.68	0.17
VQ-BET	0.10	0.27	0.12	0.13	0.06	0.64	0.59
VQ-VLA	0.14	0.46	0.24	0.30	0.20	0.59	0.93
MoEActok	**0.26**	**0.65**	0.15	**0.50**	**0.29**	**0.85**	**0.95**

Tokenizer	Place Mouse Pad	Place Container Plate	Place Phone Stand	Place Burger Fries	Shake Bottle	Average Success Rate
Binning	0.00	0.00	0.02	0.01	0.92	0.24
FAST	0.01	0.07	0.01	0.00	0.76	0.17
VQ-BET	0.00	0.54	0.04	0.15	0.87	0.29
VQ-VLA	0.08	0.79	0.23	0.52	0.96	0.45
MoEActok	**0.23**	**0.88**	**0.38**	**0.59**	**0.98**	**0.56**

Table 2. Success rates of different VLAs across Simpler-Env simulation 4 tasks.

Tokenizer	Put Spoon on Towel	Put Carrot on Plate	Stack Green on Yellow	Put Eggplant in Basket	Avg.
Bining	0.08	0.00	0.00	0.04	0.03
FAST	0.21	0.17	0.00	0.08	0.12
VQ-BET	0.04	0.04	0.00	0.00	0.02
VQ-VLA	0.29	0.33	**0.21**	0.00	0.21
MoEActok	**0.38**	**0.38**	0.13	**0.63**	**0.38**

generation with high-fidelity demonstrations and standardized evaluation protocols. The benchmark encompasses diverse manipulation tasks spanning various skill categories, including Click Bell, Place Container Plate, Pick Diverse Bottles, Move Pillbottle Pad, etc. We collect 1,600 expert demonstrations for each task, yielding a total corpus of 19,200 trajectories across 12 task categories. To enhance dataset diversity and model robustness against environmental variations, we employ domain randomization techniques during data collection following the RoboTwin [6]. We adopt the task success rate as the evaluation metric.

Simpler-Env. SimplerEnv mitigates the real-to-sim control and visual gaps inherent in simulated manipulation environments, enabling more accurate evaluation of robotic policies that closely mirror their real-world performance and behavior. We focus on four WidowX robot tasks, and they are Put Carrot on Plate, Put Spoon on Towel, Stack Green on Yellow, and Put Eggplant in Basket. Training data is sourced from the BridgeV2 dataset [29]. Following prior works, we report the trajectory success rate.

4.2. Baselines and Training Details

Baselines. We compare our MoEActok tokenizer against four representative action tokenization methods: (1) **Binning**[11] uniformly discretizes each action dimension into 256 bins per timestep; (2) **FAST**[23] applies Discrete Cosine Transform (DCT) to convert actions to the frequency domain, then uses Byte-Pair Encoding (BPE); (3) **VQ-BET**[12] employs MLPs for feature encoding followed by two-layer Residual Vector Quantization (RVQ); (4) **VQ-VLA**[31] leverages a convolutional residual VQ-VAE architecture to discretize continuous actions.

Training Details. To ensure controlled comparison, all tokenizers are integrated into an identical VLA architecture (Figure 2) with Qwen2.5-0.5B [33] as the LLM backbone. Our MoEActok uses 4 expert quantizers with codebook dimension 2048. Action chunks are configured with temporal horizons of 8 steps for RoboTwin and 4 steps for BridgeV2. Given 7-dimensional per-arm control vectors, this results in 14-dimensional inputs for bimanual RoboTwin tasks and 7-dimensional inputs for single-arm BridgeV2 tasks. Training proceeds in two stages: First, MoEActok is pre-trained using AdamW [20] with learning rate 5×10^{-5}. Second, for VLA policy learning, we freeze both MoEActok and SigLIP parameters while fine-tuning the LLM and MLP projectors using AdamW with initial learning rate 1×10^{-4} and cosine annealing. VLAs train for 10 epochs on RoboTwin and 20 epochs on BridgeV2 data.

4.3. Evaluation results on Simulation Benchmarks

RoboTwin Results. Our MoEActok-based VLA achieves the highest overall success rate of 0.56, representing a substantial improvement over competing tokenization approaches. Specifically, we observe significant performance gains compared to the binning-based VLA (0.24), FAST-based VLA (0.17), and existing VQ-VAE variants, including VQ-BET (0.29) and VQ-VLA (0.45). Our method attains superior results across the majority of evaluated tasks,

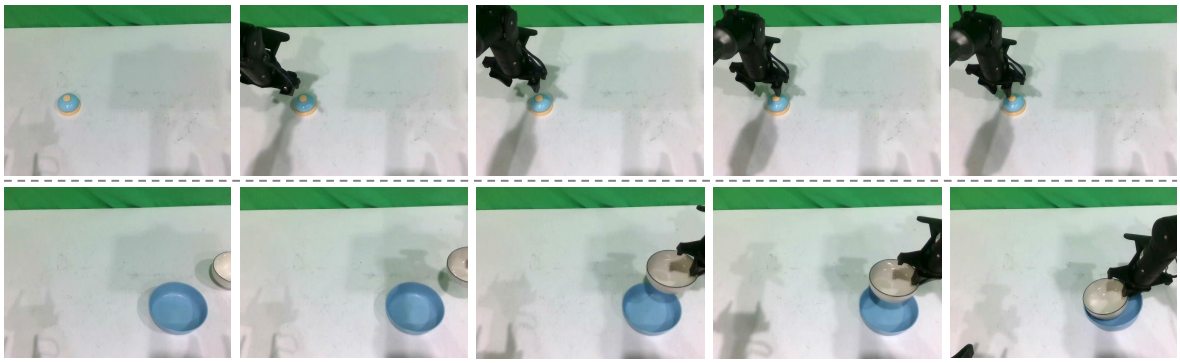


Figure 3. Real-world manipulation visualizations for "Click Bell" and "Place Container Plate" tasks.

Table 3. Results of ablation study on RoboTwin benchmark.

Tokenizer	Beat Block Hammer	Move Playingcard Away	Pick Diverse Bottles	Move Can Pot	Move Pillbottle Pad	Click Bell	Handover Mic
MoEActok	0.26	0.65	0.15	0.50	0.29	0.85	0.95
w/o Adapter	0.13	0.58	0.16	0.41	0.19	0.58	0.92
w/o Skill-aware	0.16	0.51	0.13	0.53	0.26	0.52	0.99

Tokenizer	Place Mouse Pad	Place Container Plate	Place Phone Stand	Place Burger Fries	Shake Bottle	Average Success
MoEActok	0.23	0.88	0.38	0.59	0.98	0.56
w/o Adapter	0.19	0.77	0.17	0.47	0.89	0.45
w/o Skill-aware	0.07	0.85	0.29	0.44	0.88	0.47

Table 4. Results of ablation study on Simpler-Env benchmark.

Tokenizer	Put Spoon on Towel	Put Carrot on Plate	Stack Green on Yellow	Put Eggplant in Basket	Avg.
MoEActok	0.38	0.38	0.13	0.63	0.38
w/o Adapter	0.33	0.21	0.04	0.08	0.17
w/o Skill-aware	0.25	0.33	0.29	0.33	0.30

demonstrating its robustness to diverse manipulation scenarios with varying skill requirements.

Simpler-Env Results. Table 2 presents the evaluation results on the SimplerEnv WidowX robot suite. Our MoEActok-based VLA achieves the highest average success rate of 0.38, with task-specific performance of 0.38 for Put Spoon on Towel, 0.38 for Put Carrot on Plate, 0.13 for Stack Green on Yellow, and 0.63 for Put Eggplant in Basket. Among baseline methods, VQ-VLA represents the most competitive approach, as it also employs vector quantization but within a convolutional residual architecture. The performance comparison reveals a substantial improvement: MoEActok achieves 0.38 average success rate compared to VQ-VLA's 0.21, representing an absolute gain of 17 percentage points. This significant margin demonstrates the effectiveness of our MoE quantization strategy over conventional VQ-VAE architectures. During inference, MoEActok achieves a throughput of approximately 10 Hz on a single NVIDIA RTX 4090 GPU. When integrated with the vLLM inference engine, this throughput scales to 54Hz.

4.4. Ablation Study

To systematically evaluate the contribution of each component in our MoEActok architecture, we conduct ablation studies on two critical design choices: (1) remove the adapters for latent space harmonization (**w/o Adapter**), and (2) remove the skill-aware training paradigm with explicit skill prediction (**w/o Skill-aware**). Results on RoboTwin and SimplerEnv benchmarks are presented in Tables 3 and 4, respectively. From the results, removing the adapters leads to substantial performance degradation across both benchmarks. On RoboTwin, the success rate drops from 0.56 to 0.45, while on SimplerEnv, it plummets from 0.38 to 0.17. This significant decline demonstrates that adapters play a crucial role in reconciling the representational mismatch between skill-specific quantization spaces and the shared encoder-decoder architecture. Without these harmonization modules, the model struggles to effectively leverage the specialized representations learned by each expert quantizer. The ablation of skill-aware training also yields notable performance reductions: from 0.56 to 0.47 on RoboTwin and from 0.38 to 0.30 on SimplerEnv. These results validate our hypothesis that explicit skill prediction facilitates more structured learning. By decomposing the action generation process into skill identification followed by skill-conditioned token prediction, the model can better

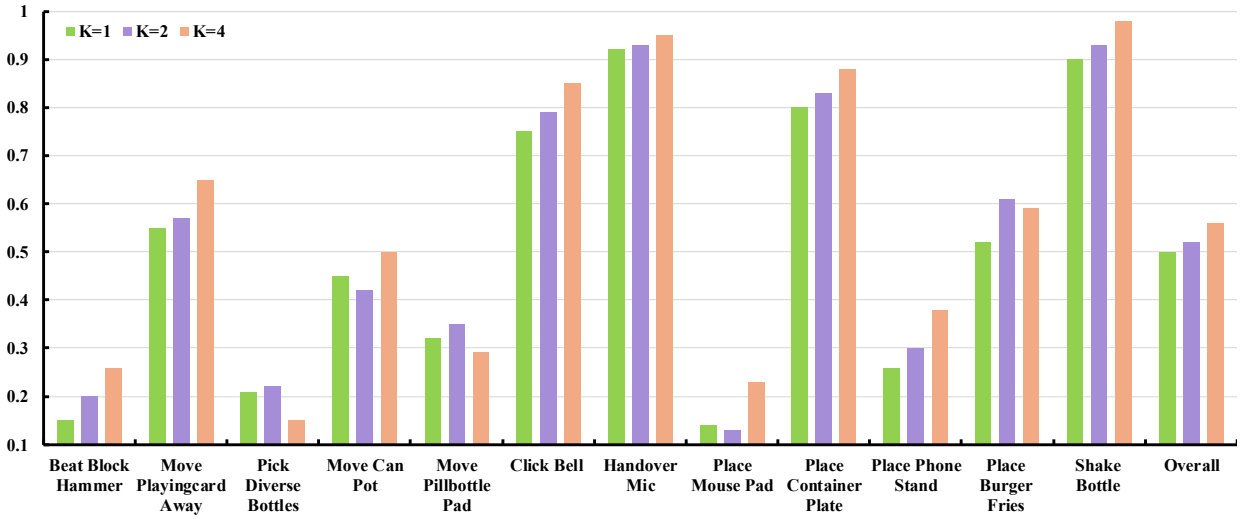


Figure 4. The influence of the number of experts on the performance of RoboTwin.

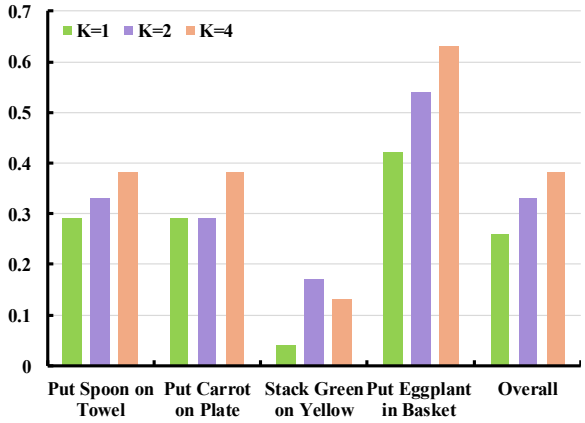


Figure 5. The influence of the number of experts on the performance of Simper-Env.

exploit the underlying skill structure in manipulation tasks, leading to more robust and generalizable policies.

4.5. In-depth Analysis

We investigate the influence of the number of expert quantizers K on performance by evaluating $K \in \{1, 2, 4\}$ on both RoboTwin and SimplerEnv benchmarks (Figures 4 and 5). The results demonstrate consistent performance gains with increasing K. On RoboTwin, the average success rate improves from 0.50 (K=1) to 0.56 (K=4). On SimplerEnv, performance increases from 0.26 (K=1) to 0.38 (K=4). These findings validate that manipulation tasks comprise diverse skill primitives requiring specialized representation.

4.6. Evaluation Results on Real Robot

To validate the practical applicability of our approach, we conduct real-world experiments on the AgileX Cobot Magic platform, equipped with four 6-DoF AgileX Piper arms with parallel grippers. Visual observations are captured via a RealSense D435 RGB camera at 640×480 resolution and 30 Hz. To evaluate sim-to-real transfer, we directly deploy VLA policies trained exclusively on RoboTwin simulation data to three real-world tasks without any fine-tuning: Click Bell, Place Container on Plate, and Pick Diverse Bottles. Each task is evaluated over 20 trials. As shown in Tab. 5, our MoEActok-based VLA significantly outperforms all baselines across all three tasks, consistent with simulation results. Fig. 3 visualizes successful execution sequences for Click Bell (top) and Place Container on Plate (bottom).

Table 5. Real-world results.

Tokenizer	Click Bell	Place Container on Plate	Pick Diverse Bottles	Avg.
Binning	6/20	0/20	0/20	0.10
FAST	4/20	1/20	0/20	0.08
VQ-BET	7/20	2/20	0/20	0.15
VQ-VLA	10/20	7/20	0/20	0.28
MoEActok	12/20	9/20	1/20	0.37

5. Conclusion

This work introduces MoEActok, a novel skill-aware action tokenizer that addresses the challenge of representing heterogeneous manipulation behaviors in VLAs. This method is motivated by the key observation that robotic manipulation tasks can be naturally decomposed into distinct skill primitives, each requiring specialized representation strategies. Based on this insight, we propose a MoE architecture that employs skill-specific quantizers within a clustering framework. This design enables MoEActok to allocate experts for learning different manipulation skills, thereby capturing the inherent diversity of action patterns more effectively. Comprehensive experiments across multiple benchmarks and real-world robotic settings validate our method.

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